

Imputation and Prediction Accuracy: New Method vs Baselines

DengueCAD hold-out experiment · NMI and NMPrediction · 03 October 2026

Protocol

- 1. valid_set.** Records without missing values (MVs) are copied aside as ground truth.
- 2. Missing values.** 20% of the valid records get 1–3 randomly chosen attributes removed (missing completely at random; the decision column is never touched). Natural MVs already in the data are kept.
- 3. Imputed dataset.** The incomplete set is imputed with the New Method Imputation (NMI) and, for comparison, MICE, kNN, MissForest and Mean/Mode. *Imputation accuracy* is the percentage of removed cells restored to their true value from valid_set.
- 4. train_set.** 30% of valid_set (stratified by class) is held out for testing; removing those records from the NMI-imputed dataset gives train_set.
- 5. Test dataset.** The held-out records are taken from valid_set and the decision (last column) is removed for 20% of them. NMPrediction, C4.5, LOR and SVM, all trained on train_set, predict those decisions. *Prediction accuracy* is the percentage matching the true valid_set decision.
- 6. Repeats.** Steps 2–5 are repeated with different random seeds (10 repeats; 3 for the 70k dengue.csv sample). Tables show mean $\pm$ standard deviation in %.

Best result in the row (ties all marked)

New Method: NMI (imputation) / NMPrediction (prediction)

Attributes are encoded as integer codes: nominal values by label, numeric values by their ordered value or decile bin when there are more than 20 distinct values. For continuous attributes an imputed cell therefore counts as correct when it falls in the true decile bin. Imputations from MICE and kNN are rounded to the nearest valid code.

1. Dengue datasets

1.1 Dengue datasets — imputation accuracy (%)

Dataset	Records	Removed cells / repeat	NMI	MICE	kNN	MissForest	Mean/Mode
dengue.csv (70k)	70,000	27,840	50.0 ± 0.2	50.1 ± 0.1	50.5 ± 0.6	50.2 ± 0.4	50.2 ± 0.0
dataset.csv (Bangladesh)	1,000	208	50.2 ± 3.6	52.2 ± 2.7	50.7 ± 2.4	49.0 ± 3.5	40.4 ± 2.8
Dengue_clinical	1,018	413	43.5 ± 1.8	45.5 ± 1.8	40.4 ± 1.8	44.3 ± 1.7	38.8 ± 1.8
Dengue_diseases (CBC)	989	376	39.1 ± 3.0	38.0 ± 2.1	37.3 ± 1.7	40.5 ± 2.2	36.2 ± 2.4
Dengue hematology	1,523	615	18.5 ± 1.4	17.7 ± 1.5	23.9 ± 1.7	36.6 ± 1.3	17.0 ± 1.1
Average			40.3	40.7	40.6	44.1	36.5
Datasets best (incl. ties)			0	2	1	2	0

dengue.csv uses its first 70,000 records. Dengue_diseases (CBC) and dataset.csv also contain natural MVs, which are imputed but not scored (they have no ground truth).

1.2 Dengue datasets — prediction accuracy (%)

Dataset	train_set	Predicted / repeat	NMPrediction	C4.5	LOR	SVM
dengue.csv (70k)	49,000	4,200	50.0 ± 0.0	49.8 ± 0.3	50.0 ± 0.1	49.7 ± 0.6
dataset.csv (Bangladesh)	847	31	99.7 ± 1.0	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0
Dengue_clinical	713	61	99.0 ± 1.6	98.2 ± 1.6	98.9 ± 0.8	99.0 ± 1.1
Dengue_diseases (CBC)	710	56	90.5 ± 2.8	85.4 ± 4.4	91.2 ± 4.2	92.1 ± 3.4
Dengue hematology	1,066	91	74.6 ± 4.3	68.5 ± 6.1	73.0 ± 2.9	75.1 ± 2.8
Average			82.8	80.4	82.6	83.2
Datasets best (incl. ties)			2	1	2	4

All four predictors are trained on the same NMI-imputed train_set. In dataset.csv the IgG test result equals the outcome, so every method can reach 100%. dengue.csv has only four binary symptoms with almost no class signal, so all methods sit near 50%.

2. Multi-disease datasets (KEEL / Weka ARFF)

2.1 Multi-disease — imputation accuracy (%)

Dataset	Records	Removed cells / repeat	NMI	MICE	kNN	MissForest	Mean/Mode
appendicitis	106	43	11.9 ± 5.6	50.0 ± 4.8	32.8 ± 9.8	40.0 ± 7.6	6.3 ± 3.8
bupa	345	143	15.5 ± 2.1	11.6 ± 2.6	12.7 ± 2.3	17.7 ± 3.1	14.5 ± 3.1
contractions	98	41	6.2 ± 3.1	29.6 ± 8.2	24.4 ± 8.1	23.8 ± 7.0	1.4 ± 1.7
heart	270	112	47.3 ± 5.3	35.2 ± 4.5	35.2 ± 5.7	45.5 ± 6.4	41.9 ± 4.1
hepatitis	155	64	58.7 ± 8.1	58.9 ± 5.4	58.2 ± 6.5	60.6 ± 6.4	58.0 ± 6.7
laryngeal1	213	89	23.6 ± 4.1	34.2 ± 3.4	32.2 ± 6.3	39.8 ± 6.1	17.9 ± 3.4
lplaudiob	3,152	1,268	48.2 ± 1.3	46.1 ± 0.7	43.9 ± 0.9	46.7 ± 1.1	41.9 ± 1.0
pima	768	312	23.7 ± 1.6	16.2 ± 2.2	17.3 ± 2.7	25.5 ± 3.3	22.7 ± 2.0
rds	85	35	34.2 ± 8.7	41.5 ± 7.7	32.8 ± 8.5	38.4 ± 7.0	33.8 ± 8.1
weaning	302	124	15.2 ± 3.9	23.2 ± 4.9	14.9 ± 3.7	26.2 ± 6.5	13.2 ± 3.1
wisconsin	699	284	58.6 ± 1.8	45.6 ± 4.1	50.2 ± 3.9	57.2 ± 2.5	52.2 ± 2.8
Average			31.2	35.6	32.2	38.3	27.6
Datasets best (incl. ties)			3	3	0	5	0

None of the ARFF files has natural MVs, so all removed cells come from step 2.

2.2 Multi-disease — prediction accuracy (%)

Dataset	train_set	Predicted / repeat	NMPrediction	C4.5	LOR	SVM
appendicitis	74	6	85.0 ± 9.5	81.7 ± 16.6	86.7 ± 10.5	86.7 ± 10.5
bupa	241	21	60.0 ± 10.8	53.8 ± 7.1	62.9 ± 7.7	64.3 ± 9.0
contractions	69	6	78.3 ± 15.8	76.7 ± 14.1	78.3 ± 11.2	88.3 ± 13.7
heart	189	16	81.2 ± 11.0	75.0 ± 9.8	81.9 ± 8.6	83.8 ± 10.3
hepatitis	109	9	83.3 ± 17.6	81.1 ± 7.5	84.4 ± 10.7	88.9 ± 9.1
laryngeal1	149	13	81.5 ± 10.4	76.2 ± 12.3	84.6 ± 12.6	86.9 ± 8.9
lplaudiob	2,206	189	61.9 ± 1.9	57.5 ± 2.5	60.8 ± 2.8	62.1 ± 2.5
pima	538	46	73.7 ± 4.9	70.4 ± 5.6	73.9 ± 4.9	73.9 ± 3.5
rds	59	5	96.0 ± 8.4	90.0 ± 10.5	96.0 ± 8.4	96.0 ± 8.4
weaning	211	18	86.7 ± 6.5	80.0 ± 8.4	85.0 ± 4.6	87.2 ± 7.4
wisconsin	489	42	94.0 ± 2.8	91.9 ± 3.9	95.5 ± 2.6	95.5 ± 3.3
Average			80.1	75.8	80.9	83.1
Datasets best (incl. ties)			1	0	4	11

Small datasets have only a handful of predicted records per repeat (for example 5 for rds), so single-repeat accuracies move in large steps; the mean over repeats is the figure to compare.

Observations

Imputation. Across all 16 datasets the best imputer counts are: NMI 3, MICE 5, kNN 1, MissForest 7, Mean/Mode 0 (ties counted for each). NMI is the best imputer on heart, lplaudiob, wisconsin, and is more accurate than Mean/Mode on 15 of 16 datasets.

Where NMI falls behind. For each incomplete record NMI treats every complete record whose distance z-score is ≤ 0 as a neighbour, i.e. all donors closer than the average distance (roughly half of them), and imputes their most frequent value. On data dominated by continuous attributes (for example appendicitis and contractions, where values are decile bins) that wide neighbourhood pulls imputations towards the overall mode, while MICE, kNN and MissForest exploit the numeric relationships between attributes. A tighter neighbour rule ($z \leq -1$, or the k closest donors) would make NMI more local on such data.

Prediction. Best predictor counts: NMPrediction 3, C4.5 1, LOR 6, SVM 15. NMPrediction is best (or tied best) on dengue.csv (70k), Dengue_clinical, rds; it beats C4.5 on 15 of 16 datasets and is on average 2.2 percentage points below the best method in each row. NMPrediction uses only the compact attribute subset chosen by the GA wrapper, while the baselines use all attributes. On small datasets one record changes accuracy by 5–20 points per repeat, so differences should be read together with the standard deviations.

Abbreviations. NMI: New Method (non-parametric) Imputation. NMPrediction: NM methodology, Algorithm 1 (NMI $\rightarrow$ GA wrapper feature selection $\rightarrow$ ADT). MICE: Multiple Imputation by Chained Equations. kNN: k-nearest-neighbour imputation ($k = 5$). MissForest: random-forest iterative imputation. C4.5: entropy decision tree. LOR: logistic regression. SVM: support vector machine (RBF kernel). ADT: Alternating Decision Tree. MCAR: missing completely at random.

Source data: results/holdout_imputation_prediction/{dengue,multi}/summary.csv and per_repeat.csv. Regenerate with scripts/run_holdout_imputation_prediction.py and scripts/build_holdout_comparison_pdf.py.